

Chaos in a three-species food chain model with a Beddington–DeAngelis functional response [☆]

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Abstract

This paper investigates a three-species food chain model with a Beddington–DeAngelis functional response, both analytically and through numerical simulations. First the equilibrium states of the system are identified and their stability analyzed analytically. The results of simulations demonstrate chaotic long-term behavior over a broad range of parameters. The existence of a strange attractor and computation of the largest Lyapunov exponent also demonstrate the presence of chaotic dynamics in the model.

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1. Introduction

Continuous time models of two interacting species have already been investigated extensively in the literature [1], and it is known that such models exhibit only steady states and limit cycles. In recent decades, however, it has been in vogue for researchers to investigate the possible existence of chaos in ecological systems. In particular, discrete-time host–parasitoid models and continuous time models with more species have been shown to exhibit complex dynamical evolution [2–10]. Multi-species food chain models have also been studied extensively, and the literature [11–17] has demonstrated that such models exhibit chaotic behavior under a Holling-type or Beddington–DeAngelis non-linear functional response [18–24]. Three-species food chain models with a ratio-dependent, Michaelis–Menten functional response have been investigated as well [25], and this model has been shown to be rich in boundary dynamics. This paper investigates the behavior of a food chain model with three trophic levels under a Beddington–DeAngelis functional response, from the theoretical and numerical perspectives.

Naji and Balasim [21] recently reported on the dynamics of such a model. Their model is described by the equations

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$$\begin{aligned}
\frac{dX}{dT} &= R_0 X \left(1 - \frac{X}{K_0} \right) - F_1(X, Y)Y, \\
\frac{dY}{dT} &= E_1 F_1(X, Y)Y - F_2(Y, Z)Z - D_1 Y, \\
\frac{dZ}{dT} &= E_2 F_2(Y, Z)Z - D_2 Z
\end{aligned} \tag{1}$$

with

$$F_i(U, V) = \frac{A_i U}{B_i V + U + C_i}, \quad i = 1, 2. \tag{2}$$

The functions $F_1(X, Y)$ and $F_2(Y, Z)$, which multiply Y and Z respectively in Eq. (1), account for interactions between two different species. The form of this functional response, given in Eq. (2), is of the Beddington–DeAngelis type. The parameters $R_0, K_0, A_i, B_i, C_i, D_i$ and E_i are all positive constants. R_0 is the intrinsic growth rate of species X , and K_0 is the carrying capacity of species X . E_1 and E_2 are the conversion rates of prey to predator for species Y and Z respectively. D_1 and D_2 are the death rates of species Y and Z respectively. This model exhibits several complex characteristics, including a steady state, limit cycles, and chaos.

In this paper we consider a different three-level trophic system with a Beddington–DeAngelis functional response. We present a model described by the following differential equations:

$$\begin{aligned}
\frac{dX}{dT} &= \frac{R_0(G_0 - X)X}{G_1 - X} - F_1(X, Y)Y, \\
\frac{dY}{dT} &= E_1 F_1(X, Y)Y - F_2(Y, Z)Z - I_1 Y, \\
\frac{dZ}{dT} &= E_2 F_2(Y, Z)Z - I_2 Z
\end{aligned} \tag{3}$$

with

$$F_i(U, V) = \frac{A_i U}{B_i V + C_i U + D_i}, \quad i = 1, 2. \tag{4}$$

The functions $F_1(X, Y)$ and $F_2(Y, Z)$ again represent a functional response of the Beddington–DeAngelis type. A_i, B_i and C_i ($i = 1, 2$) are the saturating parameters of the two responses. $\frac{A_i}{D_i}$ is the maximum harvest rate of predator from prey U , $\frac{D_i}{B_i}$ and $\frac{D_i}{C_i}$ are the half saturation constants, $i = 1, 2$. The parameters $R_0, G_0, G_1, A_i, B_i, C_i, D_i, E_i$ and I_i are all positive constants. R_0 is the intrinsic growth rate of species X , and $R_0 G_0$ is its carrying capacity. Furthermore, G_1 is the limiting value of available resources. In other words, it is the carrying capacity under ideal conditions (i.e., when no resources are wasted), which is impossible in reality. The ratio thus concerns the efficiency of nutrient utilization in species X , and has a value between zero and one. At all times, we have $X < G_0 < G_1$. When the ratio $\frac{G_0}{G_1}$ is close to one, the efficiency is high; lower values indicate that the population increment is restricted by the limiting resource [26]. E_1 and E_2 are the conversion rates of prey to predator for species Y and Z , respectively; I_1 and I_2 are the death rates of species Y and Z respectively.

Since population densities cannot be negative, the state space of system (3) is given by

$$R_+^3 = \{(X, Y, Z) \in R^3 : X \geq 0, Y \geq 0, Z \geq 0\}.$$

For easier mathematical analysis, we reduce the number of parameters by the following transformations:

$$x = \frac{X}{D_1}, \quad y = \frac{Y}{D_1}, \quad z = \frac{Z}{D}, \quad t = I_1 T. \tag{5}$$

We obtain the following dimensionless system of equations in place of Eq. (3):

$$\begin{aligned}
\frac{dx}{dt} &= \frac{r_0(k_0 - x)x}{k_1 - x} - \frac{a_1 xy}{a_2 y + a_3 x + 1} = x f(x, y, z), \\
\frac{dy}{dt} &= \frac{a_4 xy}{a_2 y + a_3 x + 1} - \frac{a_5 yz}{a_6 z + a_7 y + a_8} - y = y g(x, y, z), \\
\frac{dz}{dt} &= \frac{a_9 yz}{a_6 z + a_7 y + a_8} - a_{10} z = z h(x, y, z)
\end{aligned} \tag{6}$$

where

$$\begin{aligned} r_0 &= \frac{R_0}{I_1}, \quad k_0 = \frac{G_0}{D_1}, \quad k_1 = \frac{G_1}{D_1}, \quad a_1 = \frac{A_1}{I_1}, \quad a_2 = B_1, \quad a_3 = C_1, \\ a_4 &= \frac{A_1 E_1}{I_1}, \quad a_5 = \frac{A_2}{I_1}, \quad a_6 = B_2, \quad a_7 = C_2, \quad a_8 = \frac{D_2}{D_1}, \quad a_9 = \frac{A_2 E_2}{I_1}, \quad a_{10} = \frac{I_2}{I_1}. \end{aligned}$$

2. Stability analysis

In this section, the existence and local stability of non-negative equilibrium points in system (6) are investigated. Four non-negative equilibrium points are found in all:

- (1) The equilibrium point $E_0 = (0, 0, 0)$ always exists.
- (2) The equilibrium point $E_1 = (k_0, 0, 0)$ always exists, indicating that the prey population increases to the carrying capacity in the absence of predation.
- (3) The intermediate predator can survive in the absence of the top predator. The equilibrium point $E_2 = (x^*, y^*, 0)$ is defined by

$$y^* = \frac{x^*(a_4 - a_3) - 1}{a_2}. \quad (7)$$

For $x^* > 0$, it follows that:

$$\omega_1 > 0; \quad \omega_4 > 0; \quad x^* = \frac{\omega_2 \pm \sqrt{\omega_4}}{2\omega_1}$$

or

$$\omega_1 < 0; \quad \omega_4 > 0; \quad x^* = \frac{\omega_2 - \sqrt{\omega_4}}{2\omega_1},$$

where $\omega_1 = a_1(a_4 - a_3) - r_0 a_2 a_4$, $\omega_2 = a_1(a_4 - a_3)k_1 - r_0 a_2 a_4 k_0 + a_1$, $\omega_3 = a_1 k_1$, $\omega_4 = \omega_2^2 - 4\omega_1 \omega_3$. For $y^* > 0$, it follows that

$$x^*(a_4 - a_3) > 1.$$

- (4) A positive equilibrium point $E_3 = (\bar{x}, \bar{y}, \bar{z})$ exists if and only if there is a positive solution to the following nonlinear algebraic system:

$$\begin{cases} \frac{r_0(k_0 - \bar{x})}{k_1 - \bar{x}} - \frac{a_1 \bar{y}}{a_2 \bar{y} + a_3 \bar{x} + 1} = 0, \\ \frac{a_4 \bar{x}}{a_2 \bar{y} + a_3 \bar{x} + 1} - \frac{a_5 \bar{z}}{a_6 \bar{z} + a_7 \bar{y} + a_8} - 1 = 0, \\ \frac{a_9 \bar{y}}{a_6 \bar{z} + a_7 \bar{y} + a_8} - a_{10} = 0. \end{cases} \quad (8)$$

Eq. (8) can be rearranged to find constraints on the solution $\bar{x}, \bar{y}, \bar{z}$:

$$\bar{y} = \frac{a_{10}(a_6 \bar{z} + a_8)}{a_9 - a_7 a_{10}}, \quad \bar{z} = \frac{r_0 a_4 a_9 (a_9 - a_7 a_{10})(k_0 - \bar{x})\bar{x}}{u_1(k_1 - \bar{x})} - \frac{u_2}{u_1}. \quad (9)$$

For $\bar{x} > 0, \bar{y} > 0, \bar{z} > 0$, it follows that

$$a_9 > a_7 a_{10}; \quad u_4 > 0; \quad \frac{u_3 - \sqrt{u_4}}{2r_0 a_4 a_9 (a_9 - a_7 a_{10})} < \bar{x} < \frac{u_3 + \sqrt{u_4}}{2r_0 a_4 a_9 (a_9 - a_7 a_{10})},$$

where

$$\begin{aligned} u_1 &= a_1 a_6 a_9 a_{10} + a_1 a_5 a_{10} (a_9 - a_7 a_{10}); \quad u_2 = a_1 a_8 a_9 a_{10}, \\ u_3 &= u_2 + r_0 a_4 a_9 (a_9 - a_7 a_{10}) k_0; \quad u_4 = u_3^2 - 4r_0 a_4 a_9 (a_9 - a_7 a_{10}) u_2 k_1. \end{aligned}$$

Furthermore, we can analyze the local stability of system (6) using its Jacobian matrix at (x, y, z) :

$$J(x, y, z) = \begin{pmatrix} x \frac{\partial f}{\partial x} + f & x \frac{\partial f}{\partial y} & x \frac{\partial f}{\partial z} \\ y \frac{\partial g}{\partial x} & y \frac{\partial g}{\partial y} + g & y \frac{\partial g}{\partial z} \\ z \frac{\partial h}{\partial x} & z \frac{\partial h}{\partial y} & z \frac{\partial h}{\partial z} + h \end{pmatrix}. \quad (10)$$

(1) The Jacobian matrix of system (6) at the equilibrium point $E_0 = (0, 0, 0)$ is

$$J(0, 0, 0) = \begin{pmatrix} \frac{r_0 k_0}{k_1} & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & -a_{10} \end{pmatrix}. \quad (11)$$

The roots of the characteristic equation of $J(0, 0, 0)$ are $\lambda_1 = \frac{r_0 k_0}{k_1} > 0$, $\lambda_2 = -1 < 0$, and $\lambda_3 = -a_{10} < 0$, so $E_0 = (0, 0, 0)$ is a saddle point.

(2) The Jacobian matrix of system (6) at the equilibrium point $E_1 = (k_0, 0, 0)$ is

$$J(k_0, 0, 0) = \begin{pmatrix} -\frac{r_0 k_0}{k_1 - k_0} & -\frac{a_1 k_0}{1 + a_3 k_0} & 0 \\ 0 & \frac{a_4 k_0}{1 + a_3 k_0} - 1 & 0 \\ 0 & 0 & -a_{10} \end{pmatrix}. \quad (12)$$

The roots of the characteristic equation of $J(k_0, 0, 0)$ are $\lambda_1 = -\frac{r_0 k_0}{k_1 - k_0} < 0$, $\lambda_2 = \frac{a_4 k_0}{1 + a_3 k_0} - 1$, and $\lambda_3 = -a_{10} < 0$. $E_1 = (k_0, 0, 0)$ is thus a saddle point if $k_0(a_4 - a_3) > 1$, and a stable point if $k_0(a_4 - a_3) < 1$.

(3) The equilibrium point $E_2 = (x^*, y^*, 0)$, where x^* and y^* are given by Eq. (7), can be analyzed in terms of a subsystem of the equations in system (6). Setting $z = 0$, we obtain system (13):

$$\begin{aligned} \frac{dx}{dt} &= \frac{r_0(k_0 - x)x}{k_1 - x} - \frac{a_1 xy}{a_2 y + a_3 x + 1}, \\ \frac{dy}{dt} &= \frac{a_4 xy}{a_2 y + a_3 x + 1} - y, \end{aligned} \quad (13)$$

$E_{21} = (x^*, y^*)$ is a non-negative equilibrium point of subsystem (13). Clearly, $E_2 = (x^*, y^*, 0)$ has the same behavior as $E_{21} = (x^*, y^*)$ when the system is limited to the (x, y) plane. The Jacobian matrix of subsystem (13) at $E_{21} = (x^*, y^*)$ is

$$J(x^*, y^*) = \begin{pmatrix} C_{11} & C_{12} \\ C_{21} & C_{22} \end{pmatrix}, \quad (14)$$

where

$$\begin{aligned} G_{11} &= x^* \left[\frac{r_0(k_0 - k_1)}{(k_1 - x^*)^2} + \frac{a_1 a_3 y^*}{(a_2 y^* + a_3 x^* + 1)^2} \right]; \quad G_{12} = \frac{-a_1 x^* (a_3 x^* + 1)}{(a_2 y^* + a_3 x^* + 1)^2}; \\ G_{21} &= \frac{a_4 y^* (a_2 y^* + 1)}{(a_2 y^* + a_3 x^* + 1)^2}; \quad G_{22} = \frac{-a_2 a_4 x^* y^*}{(a_2 y^* + a_3 x^* + 1)^2}. \end{aligned}$$

The characteristic equation of J is $\lambda^2 + \delta_1 \lambda + \delta_2 = 0$, where $\delta_1 = -(G_{11} + G_{22})$, and $\delta_2 = G_{11} G_{22} - G_{12} G_{21}$. According to the Routh–Hurwitz criterion, $E_{21} = (x^*, y^*)$ is locally and asymptotically stable if and only if both δ_1 and δ_2 are greater than zero.

It is obvious that $E_2 = (x^*, y^*, 0)$ is stable in the (x, y) plane if $E_{21} = (x^*, y^*)$ is stable. However, the roots of the characteristic equation of the Jacobian matrix of system (6) at this equilibrium point satisfy

$$\begin{aligned} \xi_1 + \xi_2 &= x^* \left[\frac{r_0(k_0 - k_1)}{(k_1 - x^*)^2} + \frac{(a_1 a_3 - a_2 a_4) y^*}{(a_2 y^* + a_3 x^* + 1)^2} \right], \\ \xi_1 \xi_2 &= -\frac{r_0(k_0 - k_1) a_2 x^* y^*}{(k_1 - x^*)^2 (a_2 y^* + a_3 x^* + 1)} + \frac{a_1 y^*}{(a_2 y^* + a_3 x^* + 1)^2}, \\ \xi_3 &= \frac{a_9 y^*}{a_7 y^* + a_8} - a_{10}. \end{aligned} \quad (15)$$

The stability of $E_2 = (x^*, y^*, 0)$ in the positive z direction depends on ξ_3 . $E_2 = (x^*, y^*, 0)$ is stable in the positive z direction if $\xi_3 < 0$.

(4) The Jacobian matrix of system (6) at the equilibrium point

$E_3 = (\bar{x}, \bar{y}, \bar{z})$ is

$$J(\bar{x}, \bar{y}, \bar{z}) = \begin{pmatrix} H_{11} & H_{12} & H_{13} \\ H_{21} & H_{22} & H_{23} \\ H_{31} & H_{32} & H_{33} \end{pmatrix}, \quad (16)$$

where

$$H_{11} = \bar{x} \left[\frac{r_0(k_0 - k_1)}{(k_1 - \bar{x})^2} + \frac{a_1 a_3 \bar{y}}{(a_2 \bar{y} + a_3 \bar{x} + 1)^2} \right];$$

$$H_{12} = \frac{-a_1 \bar{x}(a_3 \bar{x} + 1)}{(a_2 \bar{y} + a_3 \bar{x} + 1)^2} < 0;$$

$$H_{13} = 0;$$

$$H_{21} = \frac{a_4 \bar{y}(a_2 \bar{y} + 1)}{(a_2 \bar{y} + a_3 \bar{x} + 1)^2} > 0;$$

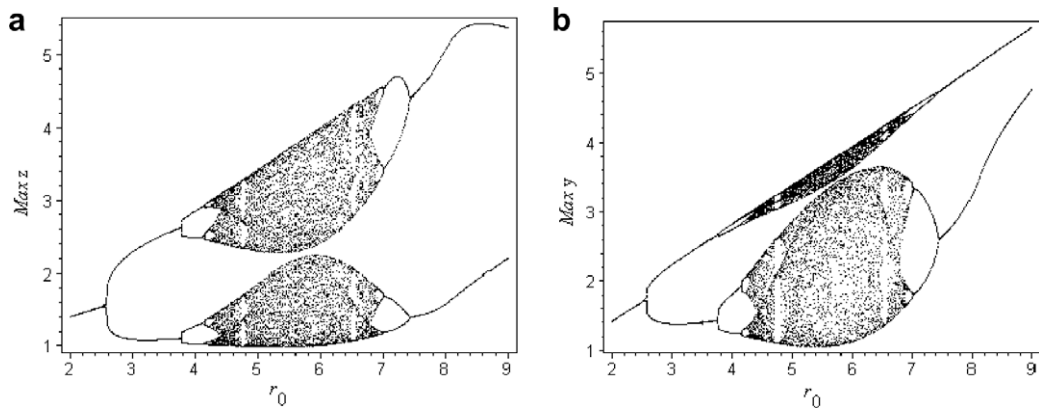


Fig. 1. Bifurcation diagram of the system (6) with $k_0 = 3$, $k_1 = 4$, $a_1 = 4$, $a_2 = 0.2$, $a_3 = 0.8$, $a_4 = 3$, $a_5 = 2$, $a_6 = 0.15$, $a_7 = 0.8$, $a_8 = 0.5$, $a_9 = 1.5$, $a_{10} = 0.2$, $2 < r_0 < 9$, and initial conditions $x_0 = (0.1, 0.1, 0.1)$: (a) maxima for top predator z ; (b) maxima for intermediate predator y .

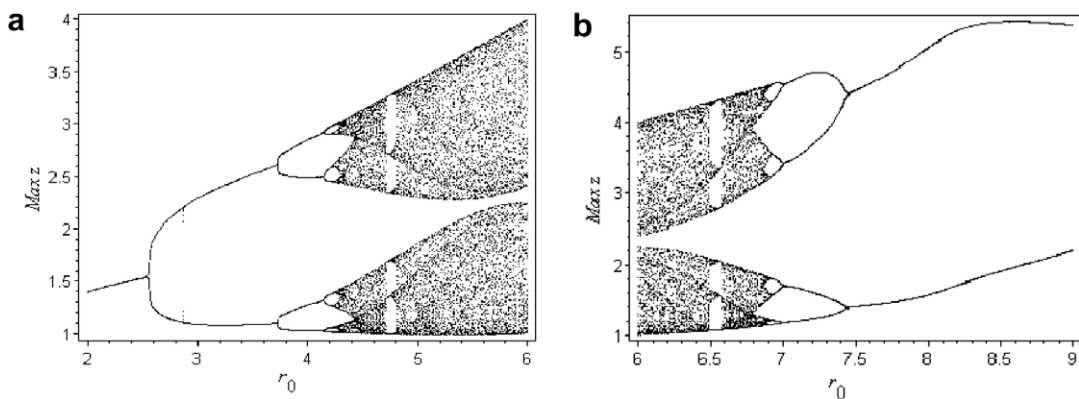


Fig. 2. Magnification of Fig. 1a: (a) $2 < r_0 < 6$; (b) $6 < r_0 < 9$.

$$H_{22} = \bar{y} \left[\frac{-a_2 a_4 \bar{x}}{(a_2 \bar{y} + a_3 \bar{x} + 1)^2} + \frac{a_5 a_7 \bar{z}}{(a_6 \bar{z} + a_7 \bar{y} + a_8)^2} \right];$$

$$H_{23} = \frac{-a_5 \bar{y} (a_7 \bar{y} + a_8)}{(a_6 \bar{z} + a_7 \bar{y} + a_8)^2} < 0;$$

$$H_{31} = 0;$$

$$H_{32} = \frac{a_9 \bar{z} (a_6 \bar{z} + a_8)}{(a_6 \bar{z} + a_7 \bar{y} + a_8)^2} > 0;$$

$$H_{33} = -\frac{a_6 a_9 \bar{z} \bar{y}}{(a_6 \bar{z} + a_7 \bar{y} + a_8)^2} < 0.$$

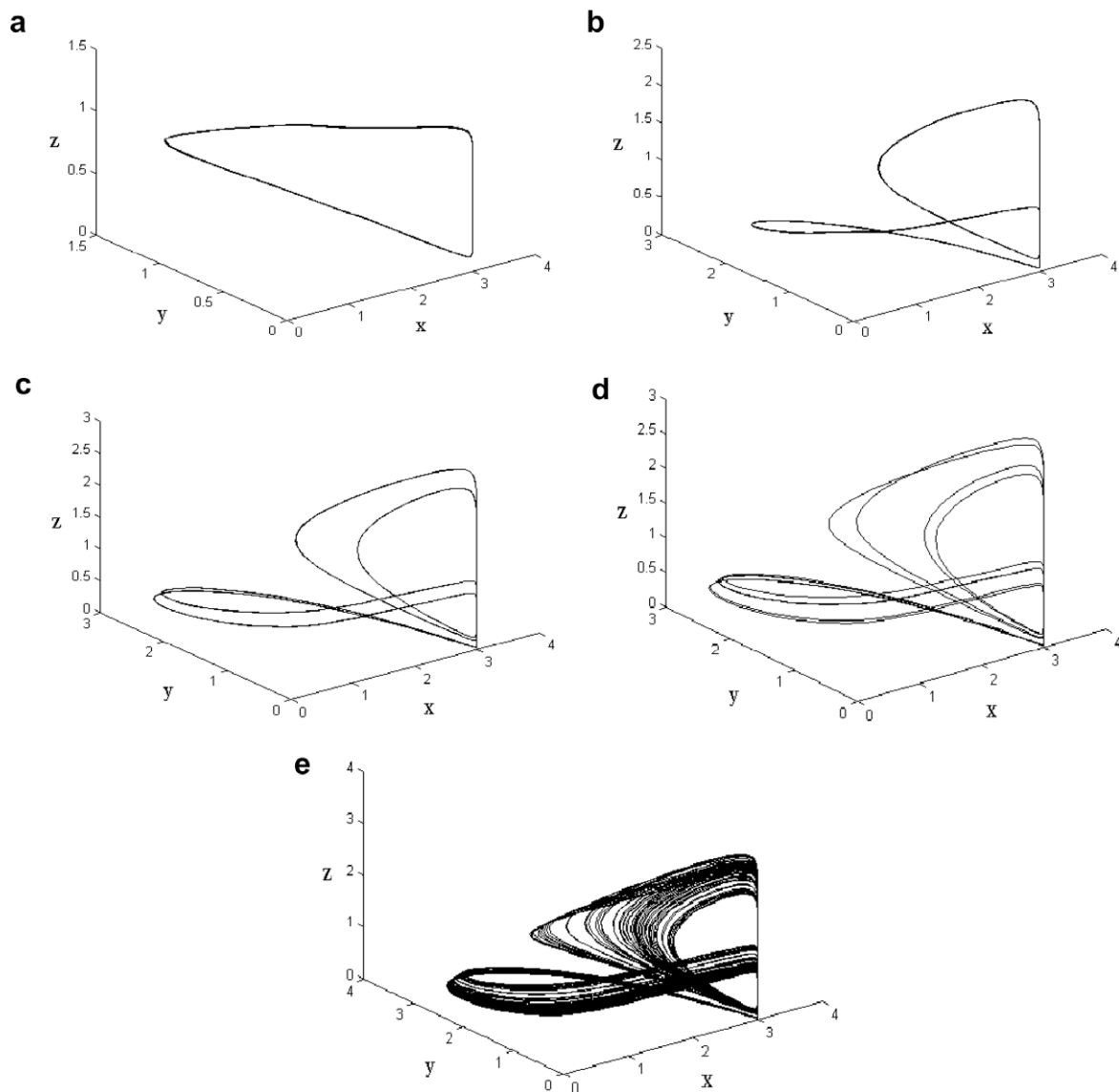


Fig. 3. Period-doubling bifurcation: (a) period-1 attractors appear when $r_0 = 2.1$; (b) period-2 attractors appear when $r_0 = 3$; (c) period-4 attractors appear when $r_0 = 3.9$; (d) period-8 attractors appear when $r_0 = 4.2$; (e) chaos occurs when $r_0 = 4.5$.

The characteristic equation of $J(\bar{x}, \bar{y}, \bar{z})$ is $\lambda^3 + \sigma_1 \lambda^2 + \sigma_2 \lambda + \sigma_3 = 0$, with the coefficients

$$\sigma_1 = -(H_{11} + H_{22} + H_{33}),$$

$$\sigma_2 = H_{11}H_{22} - H_{12}H_{21} + H_{22}H_{33} - H_{23}H_{32} + H_{11}H_{33},$$

$$\sigma_3 = H_{11}H_{23}H_{32} - H_{11}H_{22}H_{33} + H_{12}H_{21}H_{33}.$$

According to the Routh–Hurwitz criterion, $E_3 = (\bar{x}, \bar{y}, \bar{z})$ is locally and asymptotically stable if

$$\sigma_1 > 0, \quad \sigma_3 > 0, \quad \sigma_1 \sigma_2 > \sigma_3. \quad (17)$$

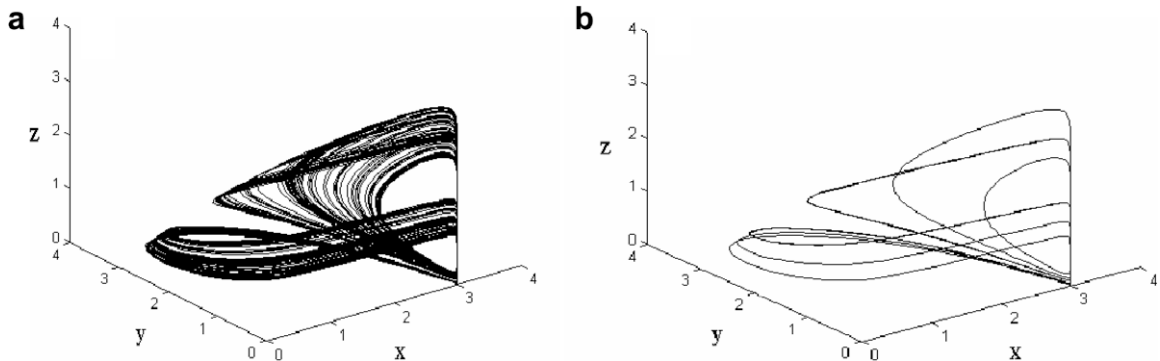


Fig. 4. Crisis is shown: When the parameter r_0 is increased slightly beyond $r_0 = 4.7$, the chaos suddenly disappears: (a) chaos occurs when $r_0 = 4.7$; (b) period-6 attractors appear when $r_0 = 4.75$.

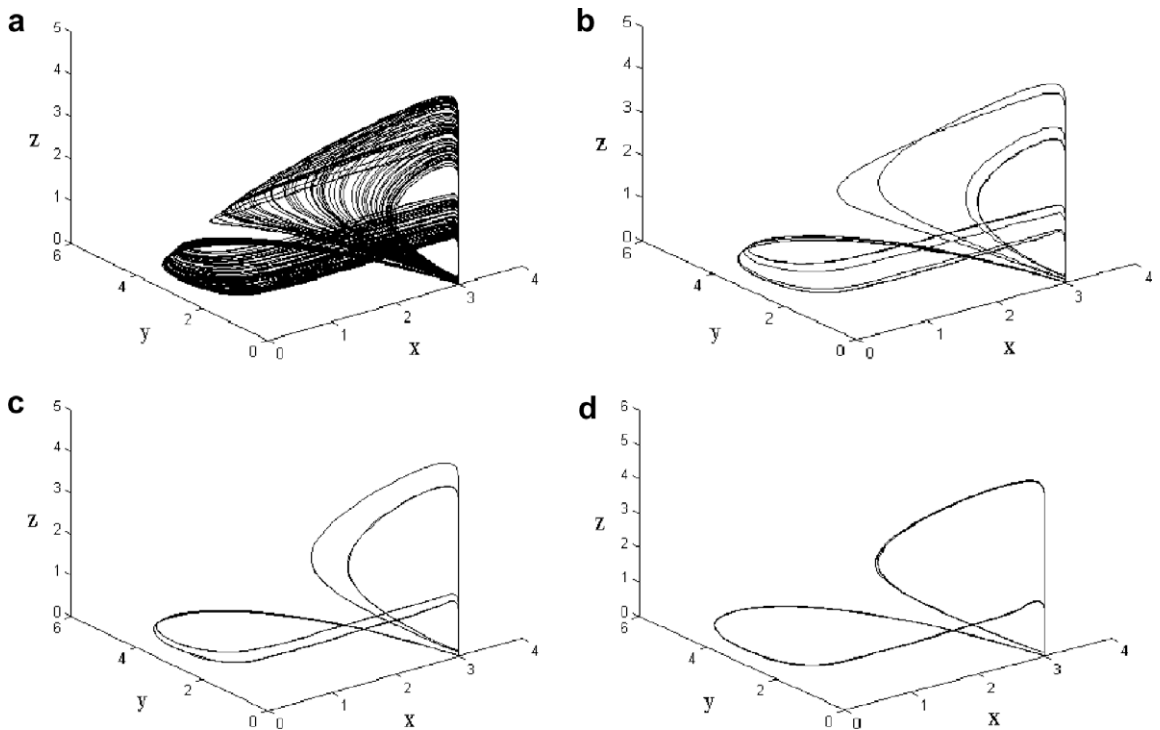


Fig. 5. Period-halving bifurcation: (a) chaos occurs when $r_0 = 6.7$; (b) period-8 attractors appear when $r_0 = 7$; (c) period-4 attractors appear when $r_0 = 7.4$; (d) period-2 attractors appear when $r_0 = 8$.

3. Numerical simulations

In this section, the global dynamical behavior of system (6) is investigated using numerical simulations. As we are mainly interested in the long-term behavior of system (6), we plot only the maximum values of the various time series. The numerical simulations were implemented in Maple. A variety of parameters are investigated, in particular the intrinsic growth rate r_0 and the carrying capacities k_0 and k_1 of species x .

Fig. 1 shows the bifurcation diagram of system (6) for successive maxima of the intermediate predator population y and the top predator population z . The initial population values are $x_0 = (x_0, y_0, z_0) = (0.1, 0.1, 0.1)$. Because of the similarity of the two bifurcation diagrams, only Fig. 1a is magnified in Fig. 2 and we only analyze the long-term behavior of system (6) in terms of Fig. 1a. Here, successive maxima for the top predator population z are plotted as a function of the bifurcation parameter r_0 .

As r_0 increases from 2 to 2.6, z rapidly grows and a stable limit cycle is observed. A period-doubling bifurcation occurs at $r_0 = 2.6$; at this point the period-1 attractor changes to a period-2 attractor. As r_0 increases further, it doubles again to a period-4 attractor, and again, until eventually the period-doubling bifurcations lead the system into a chaotic state (Fig. 3). The bifurcation diagram in the range $[4.3, 6.8]$ shows many obvious complexities, such as chaotic bands with periodic windows, tangent bifurcations, period-doubling bifurcations, an attractor crisis (Fig. 4), and chaotic attractors. For $r_0 > 6.7$, however, chaotic behavior disappears after a sequence of period-halving bifurcations (Fig. 5).

Fig. 6 shows the bifurcation diagram of system (6) for successive maxima of the intermediate predator population y and the top predator population z . The initial values are $x_0 = (0.1, 0.1, 0.1)$. Because of the similarity of the two bifurcation diagrams, only Fig. 6a is magnified in Fig. 7 and we analyze the long-term behavior of the system in terms of

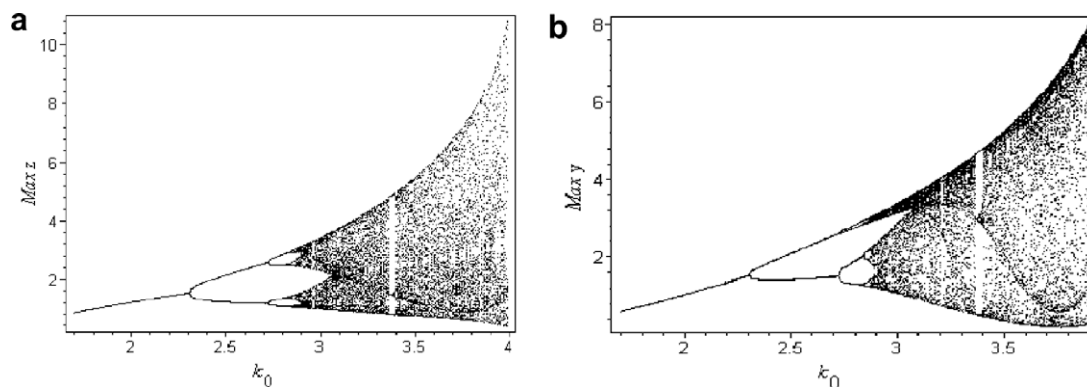


Fig. 6. Bifurcation diagram of system (6) with $r_0 = 5$, $k_1 = 4$, $a_1 = 4$, $a_2 = 0.2$, $a_3 = 0.8$, $a_4 = 3$, $a_5 = 2$, $a_6 = 0.15$, $a_7 = 0.8$, $a_8 = 0.5$, $a_9 = 1.5$, $a_{10} = 0.2$, $1.7 < k_0 < 4$, and initial conditions $x_0 = (0.1, 0.1, 0.1)$: (a) maxima for the top predator population z ; (b) maxima for the intermediate predator population y .

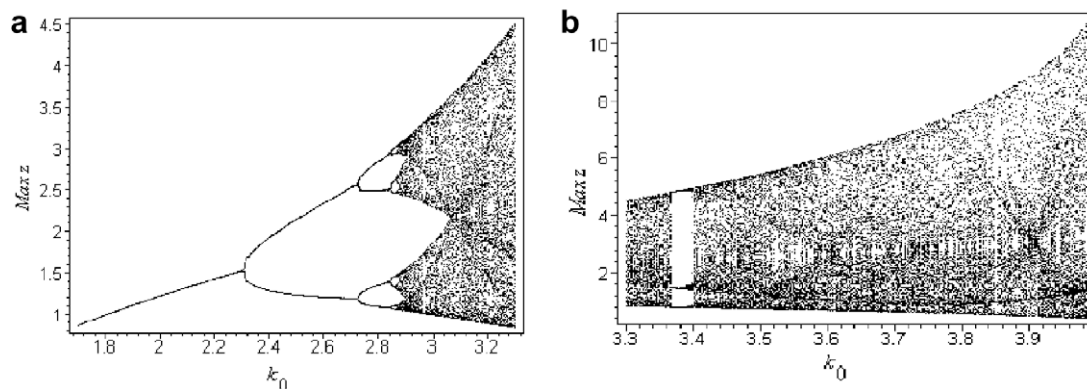


Fig. 7. Magnification of Fig. 6a: (a) $1.7 < k_0 < 3.3$; (b) $3.3 < k_0 < 4$.

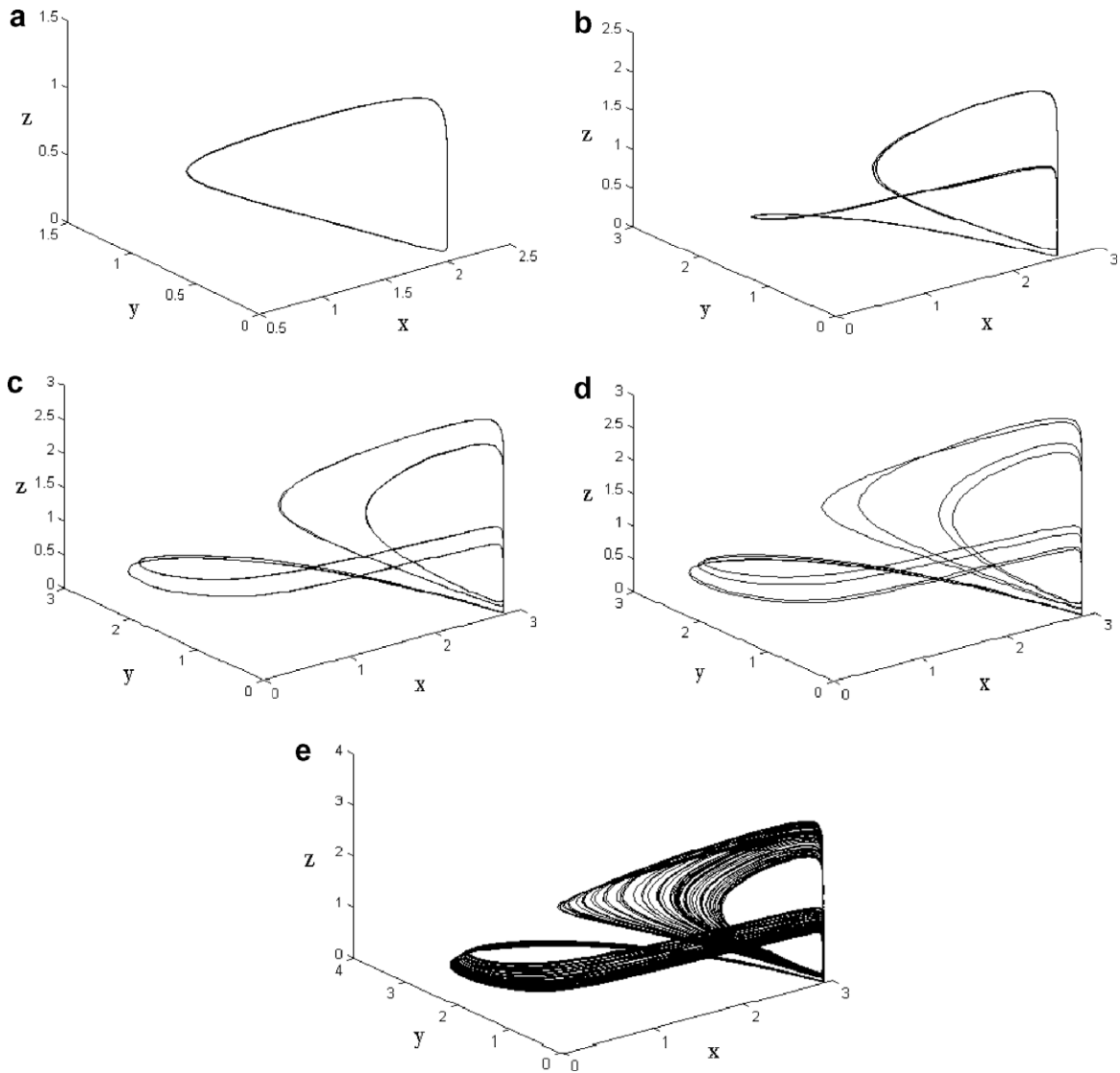


Fig. 8. Period-doubling bifurcation: (a) period-1 attractors appear when $k_0 = 2$; (b) period-2 attractors appear when $k_0 = 2.5$; (c) period-4 attractors appear when $k_0 = 2.8$; (d) period-8 attractors appear when $k_0 = 2.85$; (e) chaos occurs when $k_0 = 2.9$.

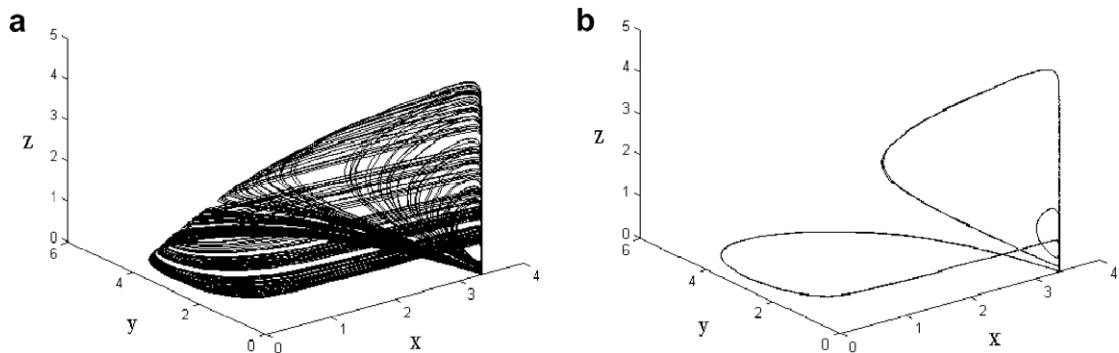


Fig. 9. Crisis is shown: When the parameter k_0 is slightly increased beyond $k_0 = 3.35$, the chaos suddenly disappears: (a) chaos when $k_0 = 3.35$; (b) period-3 attractors when $k_0 = 3.38$.

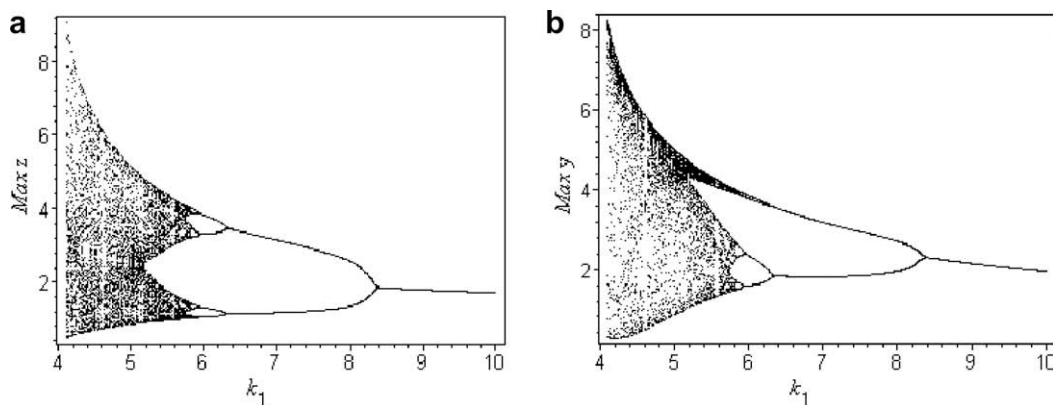


Fig. 10. Bifurcation diagram of the system (6) with $r_0 = 5$, $k_0 = 4$, $a_1 = 4$, $a_2 = 0.2$, $a_3 = 0.8$, $a_4 = 3$, $a_5 = 2$, $a_6 = 0.15$, $a_7 = 0.8$, $a_8 = 0.5$, $a_9 = 1.5$, $a_{10} = 0.2$, $4.1 < k_1 < 10$, and initial conditions $x_0 = (0.1, 0.1, 0.1)$: (a) maxima for the top predator population z ; (b) maxima for the intermediate predator population y .

Fig. 6a. Here, successive maxima of the top predator population z are plotted as a function of the bifurcation parameter k_0 . Values of k_0 between 1.7 and 2.3 result in a stable limit cycle, which again is followed by a cascade of period-doubling bifurcations (Fig. 8) as k_0 increases. This behavior eventually leads the system into a chaotic band, with features such as periodic windows, tangent bifurcations, period-halving bifurcations, period-doubling bifurcations, an attractor crisis (Fig. 9), and chaotic attractors.

Adjusting the parameter k_1 can easily destabilize a steady state, thanks to the complex dynamics of system (6). Bifurcation diagrams (Fig. 10) are shown for system (6) with k_1 as the bifurcation parameter. Varying k_1 will cause the system to fall into one of two regions: the chaotic region (4–5.6), or the region of limit cycles (5.6–10). In the latter case, the system eventually enters a stable state after experiencing a cascade of period-halving bifurcations.

4. The largest Lyapunov exponent

Convincing evidence for deterministic chaos has come from a variety of recent experiments on dissipative non-linear systems [27–29]. Therefore, the problem of detecting and quantifying chaos has become an important one. Here the largest Lyapunov exponent is considered, which has proven to be the most useful dynamical diagnostics for chaotic system. It is the average exponential rates of divergence or convergence of nearby orbits in phase space [27]. The general principle for calculating the largest Lyapunov exponent is to follow two nearby orbits and to calculate their average

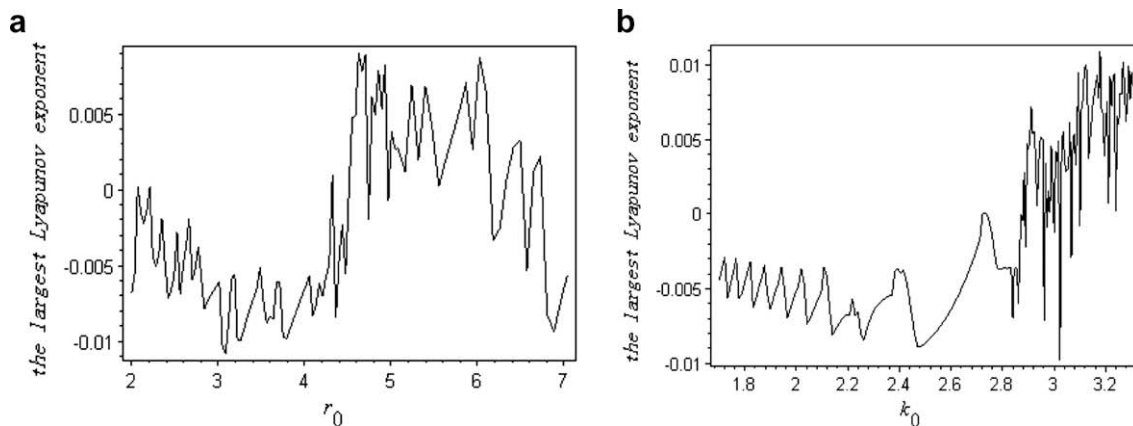


Fig. 11. (a) The largest Lyapunov exponent (r_0 ranging from 2 to 7.04) for system (6); (b) The largest Lyapunov exponent (k_0 ranging from 1.7 to 3.30) for system (6).

logarithmic rate of separation. Whenever they move too far apart, one of the orbits has to be moved back to the vicinity of the other along the line of separation. The current approach is principally based on the work of Sprott [28].

For a chaotic attractor, the largest Lyapunov exponent λ must be positive. If λ is negative, this implies a stability state or a period attractor. Reviewing the bifurcation diagram for the top predator in Fig. 1a and Fig. 6a, we can calculate the corresponding largest Lyapunov exponent (r_0 from 2 to 7.04, k_0 from 1.7 to 3.30) for system (6). The output is shown in Fig. 11a and b.

5. Conclusion

In this paper we have examined the dynamic complexities of a three-species food chain with a Beddington–DeAngelis functional response. Many complexities are observed, such as chaotic bands with periodic windows, tangent bifurcations, period-doubling bifurcations, period-halving bifurcations, and chaotic attractors. Furthermore, by means of computer simulations we can use the largest Lyapunov exponent to detect and quantify chaos.

References

- [1] Kuang Y, Beretta E. Global quantities analysis of a ratio-dependent predator–prey system. *J Math Biol* 1998;36:389–406.
- [2] Tang SY, Chen LS. Chaos in functional response host–parasitoid ecosystem models. *Chaos Solitons & Fractals* 2002;13:875–84.
- [3] Lv S, Zhao M. The dynamic complexity of a host–parasitoid model with a lower bound for the host. *Chaos Solitons & Fractals* 2008;36:911–9.
- [4] Hui J, Zhu D. Dynamic complexities for prey-dependent consumption integrated pest management models with impulsive effects. *Chaos Solitons & Fractals* 2006;29:233–51.
- [5] Sun M, Tian L, Fu Y, Qian W. Dynamics and adaptive synchronization of the energy resource system. *Chaos Solitons & Fractals* 2007;31:879–88.
- [6] Letellier C, Aziz-Alaoui MA. Analysis of the dynamics of a realistic ecological model. *Chaos Solitons & Fractals* 2002;13:95–107.
- [7] Sheu LJ, Chen HK, Chen JH, Tam LM. Chaos in a new system with fractional order. *Chaos Solitons & Fractals* 2007;312:1203–12.
- [8] Hastings A, Powell T. Chaos in a three-species food chain. *Ecology* 1991;72:896–903.
- [9] Gakkhar S, Naji RK. Order and chaos in predator to prey ratio-dependent food chain. *Chaos Solitons & Fractals* 2003;18:229–39.
- [10] Gakkhar S, Naji RK. Chaos in three species ratio dependent food chain. *Chaos Solitons & Fractals* 2002;14:771–8.
- [11] Sabin GCW, Sumner D. Chaos in a periodically forced predator–prey ecosystem model. *Math Biosci* 1993;113:91–113.
- [12] Vandermmeer J, Maruca S. Indirect effects with a keystone predator: coexistence and chaos. *Theor Popul Biol* 1998;54:38–43.
- [13] Hui J, Chen L. Dynamic complexities in a periodically pulsed ratio-dependent predator–prey ecosystem modeled on a chemostat. *Chaos Solitons & Fractals* 2006;29:407–16.
- [14] Rai V, Kumar Upadhyay R. Chaotic population dynamics and biology of the top-predator. *Chaos Solitons & Fractals* 2004;21:1195–204.
- [15] Peet AB, Deutsch PA, Peacock-Lopez E. Complex dynamics in a three-level trophic system with intraspecies interaction. *J Theor Biol* 2005;232:491–503.
- [16] Kumar Upadhyay R, Iyengar SRK. Effect of seasonality on the dynamics of 2 and 3 species prey–predator systems. *Nonlinear Anal Real World Appl* 2005;6:509–30.
- [17] Lv S, Zhao M. The dynamic complexity of a three species food chain model. *Chaos Solitons & Fractals* 2008;37:1469–80.
- [18] Li Z, Wang W, Wang H. The dynamics of a Beddington-type system with impulsive control strategy. *Chaos Solitons & Fractals* 2006;29:1229–39.
- [19] Cantrell RS, Cosner C. On the dynamics of predator–prey models with the Beddington–DeAngelis functional response. *J Math Anal Appl* 2001;257:206–22.
- [20] Gakkhar S, Naji RK. Seasonality perturbed prey–predator system with predator-dependent functional response. *Chaos Solitons & Fractals* 2003;18:1075–83.
- [21] Naji RK, Balasim AT. Dynamical behavior of a three species food chain model with Beddington–DeAngelis functional response. *Chaos Solitons & Fractals* 2007;32:1853–66.
- [22] Hwang TW. Global analysis of the predator–prey system with Beddington–DeAngelis functional response. *J Math Anal Appl* 2003;281:395–401.
- [23] Hwang TW. Uniqueness of limit cycles of the predator–prey system with the Beddington–DeAngelis functional response. *J Math Anal Appl* 2004;290:113–22.
- [24] Zhang S, Tan D, Chen L. Chaotic behavior of a chemostat model with Beddington–DeAngelis functional response and periodically impulsive invasion. *Chaos Solitons & Fractals* 2006;29:474–82.
- [25] Beddington JR. Mutual interference between parasites or predators and its effect on searching efficiency. *J Animal Ecol* 1975;44:331–40.

- [26] Qiwu C, Lawson GJ. Study on models of single populations: an expansion of the logistic and exponential equations. *J Theor Biol* 1982;98:645–59.
- [27] Rosenstein MT, Collins JJ, De Luca CJ. A practical method for calculating largest Lyapunov exponents from small data sets. *Physica D* 1993;65:117–34.
- [28] Sprott JC. *Chaos and time-series analysis*. Oxford: Oxford University Press; 2003. p. 116–7.
- [29] Grond F, Diebner HH, Sahle S, Mathias A. A robust, locally interpretable algorithm for Lyapunov exponents. *Chaos Solitons & Fractals* 2003;16:841–52.